

Unit 5, Lesson 9: Converting Units of Volume

Benchmark

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

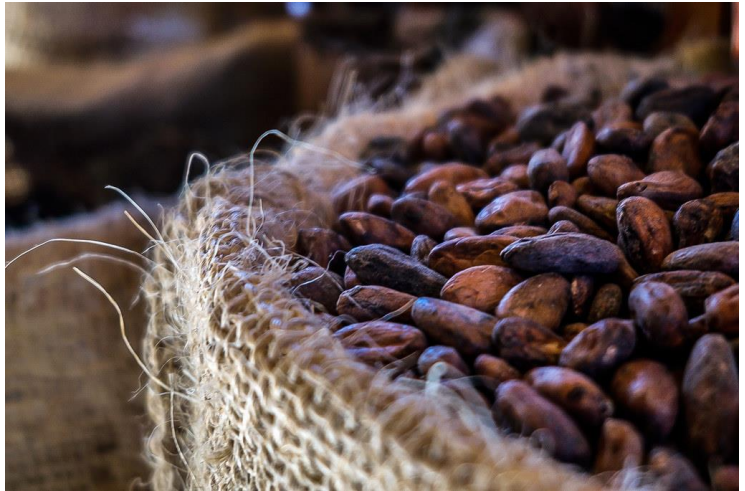
Learning Target

- ☐ I can convert liquid measurements and volume across different measurement systems.

5.9.1 Warm-Up: Mayan Currency

The ancient Maya used cacao beans – the main ingredient in chocolate – as a form of currency.

They did not use coins as money, but instead mostly bartered and traded items such as tobacco, maize (corn), and clothing. Archeologists have found that chocolate was used as currency by the 8th century CE, in the form of fermented and dried cacao beans. Maya artwork indicates that cacao beans could be exchanged for goods and services, including payment for labor and paying taxes.



Cacao, just like any other currency, could be converted to other kinds of money, including Spanish gold and silver coins such as the real. It has been found that 200 cacao beans were worth 1 Spanish real.

1. If 24,000 cacao beans was worth 15 pesos, what was the rate of cacao beans for 1 peso?
2. How many pesos were equal to 1 Spanish real?

5.9.2 Activity: Pouring Liquids to Find Conversion Rates

Each of us uses containers that have volume. Most of those containers are labeled with both U.S. customary measurements and liquid metric measurements.

1. List some U.S. customary measurements and liquid metric measurements.

Many soft drinks come in 2-liter bottles, while milk is often sold by the gallon.

2-Liter Soda Bottle	1 Gallon Milk Container
	

2. Which container do you think holds more liquid? Explain your reasoning.

Both liters (L) and gallons (gal) are units used to measure liquid volume. Liters are units within the metric system, while gallons are part of the U.S. customary system.

3. What do you predict is the relationship between liters and gallons based on the size of the containers? Write your prediction under the initial prediction heading in the table below.

Initial Prediction	Revised Prediction
___ gallon(s) $\approx$ ___ liter(s)	___ gallon(s) $\approx$ ___ liter(s)

Watch the short video of liquid from one container being poured into the other.



4. Based on your initial prediction and the amount of liquid in each container now, discuss the relationship between liters and gallons. Then, write a revised prediction in the table above.

Continue watching the remainder of the video.

5. Based on the video, what is your final estimate for the relationship between liters and gallons?

Final Estimate
___ gallon(s) $\approx$ ___ liter(s)

6. How close is your estimate to the actual conversion rate?

The big reveal ...



Actual Conversion Rate
_____ gallon(s) $\approx$ _____ liter(s)

5.9.3 Try It: Converting Liquid Volume Measurements

1. Aleah is mixing concrete for a home-improvement project. According to the bag directions, she needs 6 gallons (gal) of water to mix the amount of concrete needed. The mixing bucket only has markings in liters (L). How many liters of water does she need to measure out for mixing the concrete? Round to the nearest tenth.
2. Most cars have a fuel tank that can hold between 45 and 65 liters (L) of gasoline. What is the range of gasoline a typical fuel tank can hold in terms of gallons (gal)? Round to the nearest hundredth.

5.9.4 Guided Practice: Converting Liquid Volume Measurements

The units commonly used for liquid capacity within each system are shown, listed from smallest to largest.

U.S. Customary System	Metric System
fluid ounces (fl oz) cups (c) pints (p) quarts (qt) gallons (gal)	milliliters (mL) liters (L)

The tables below show the conversion rates for common units of liquid capacity in the U.S. customary and metric systems.

U.S. Customary System to Metric System	
1 fluid ounce (fl oz)	= 28.413 milliliters (mL)
1 cup (c)	= 236.588 milliliters (mL)
1 fluid ounce (fl oz)	= 0.030 liters (L)
1 cup (c)	= 0.24 liters (L)
1 quart (qt)	= 0.946 liters (L)
___ gallon (gal)	= _____ liters (L)

Metric System to U.S. Customary System	
1 milliliter (mL)	= 0.034 fluid ounces (fl oz)
1 milliliter (mL)	= 0.004 cups (c)
1 liter (L)	= 33.814 fluid ounces (fl oz)
1 liter (L)	= 4.23 cups (c)
1 liter (L)	= 1.057 quarts (qt)
___ liter (L)	= _____ gallons (gal)

- Use the conversion rates to find each value below. Round to the nearest hundredth.
 - 240 mL $\approx$ _____ c
 - 100 fl oz $\approx$ _____ L
- There are 100 centiliters (cL) in a liter. How many centiliters are in 2.75 quarts?

5.9.5 Guided Instruction: Cubic Units of Volume

Volume of solid figures is measured in cubic units. In some settings, liquid volume is also measured in cubic units. For instance, in many medical situations, medicines and other fluids are measured as cubic centimeters.

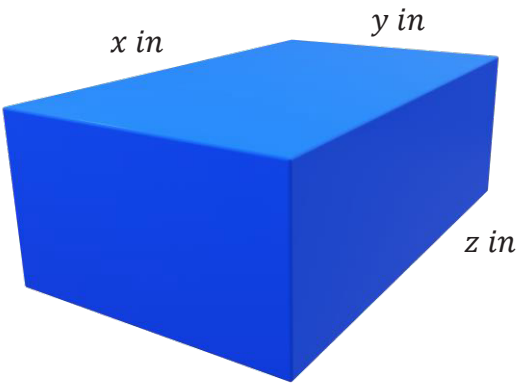
1 cubic centimeter = 1 milliliter

1. Complete the equation. Show your work and round to the nearest hundredth.

4.5 fl oz ≈ _____ cubic centimeters (cm^3)

For solid volume, the conversion rates can be determined using the conversion rates for length.

There are two ways to find the volume of a rectangular prism in cubic centimeters when the prism’s side measurements are given in inches.

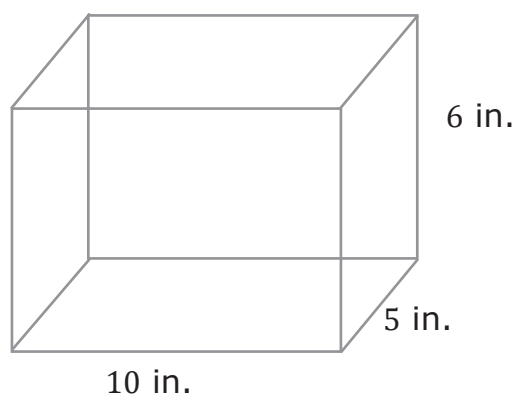


Method 1	Method 2
Convert each side length to centimeters and then calculate the volume of the figure in cubic centimeters.	Calculate the volume of the figure in cubic inches and then convert the volume to cubic centimeters.

2. Work with your partner to determine a conversion factor that can be used to convert cubic inches (in^3) to cubic centimeters (cm^3). Round to the nearest thousandth. Use the known conversion of 1 inch (in.) $\approx$ 2.54 centimeters (cm).

____ cubic inch (in^3) $\approx$ ____ cubic centimeters (cm^3)

3. How many cubic centimeters can the rectangular prism with dimensions of 10 in. by 5 in. by 6 in. hold?



5.9.6 Your Turn!

1. Use conversion rates to find each value. Round to the nearest thousandth.
 - a. 6 cups $\approx$ _____ milliliters
 - b. 300 milliliters $\approx$ _____ fluid ounces
 - c. 2 liters $\approx$ _____ quarts
 - d. 10 gallons $\approx$ _____ liters
2. Mateo buys a package with 12 bottles of water. Each bottle holds 500 milliliters of water. How many fluid ounces are in the package?

5.9.7 Wrap-Up: Ticket Out the Door

1. For each pair of liquid measurements, circle the conversion factor to use to convert from the measurement on the left to the measurement on the right.
- | Conversion Table | | | |
|--------------------|---|---|-------------|
| Converting from... | | | ...to |
| fluid ounces | $\frac{1\text{ fl oz}}{28.413\text{ mL}}$ | $\frac{28.413\text{ mL}}{1\text{ fl oz}}$ | milliliters |
| milliliters | $\frac{0.004\text{ c}}{1\text{ mL}}$ | $\frac{283.588\text{ mL}}{1\text{ c}}$ | cups |
| liters | $\frac{1.057\text{ qt}}{1\text{ L}}$ | $\frac{1\text{ L}}{1.057\text{ qt}}$ | quarts |
| gallons | $\frac{1\text{ gal.}}{3.785\text{ L}}$ | $\frac{1\text{ L}}{0.264\text{ gal.}}$ | liters |
2. Carmita is mixing sweet tea and lemonade to make her favorite beverage. She mixes 20 fluid ounces (fl oz) of sweet tea with 16 fluid ounces of lemonade. How many liters (L) of the beverage did she mix?